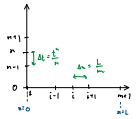
$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$ [Unsteady 1D, conduction]

1) Discretize time & space domain $T_i^n = T(x_i, t^n)$



Need m equations to solve for value of T at every grid point

Transforming PDE to FDEs (finite difference equation): We use the Taylor expansion

$$f(x+\Delta x, y) = f(x,y) + \frac{\partial f}{\partial x}\Delta x + \frac{\partial^2 f}{\partial x^2}\frac{\Delta x^2}{2!} + \frac{\partial^3 f}{\partial x^3}\frac{\Delta x^3}{3!} + \cdots$$

$$f(x-\Delta x, y) = f(x,y) - \frac{\partial f}{\partial x}\Delta x + \frac{\partial^2 f}{\partial x^2}\frac{\Delta x^2}{2!} - \frac{\partial^3 f}{\partial x^3}\frac{\Delta x^3}{3!} + \cdots$$

$$\therefore \underbrace{\frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}}_{\text{difference operator}} = \underbrace{\frac{\partial f}{\partial x}}_{\text{derivative}} + \frac{1}{2\Delta x}\underbrace{\left[\frac{2\Delta x^3}{3!}\frac{\partial^3 f}{\partial x^3} + \frac{2\Delta x^5}{5!}\frac{\partial^5 f}{\partial x^5} + \cdots\right]}_{\text{truncation error (TE)}}$$

$\mathcal{O}(\Delta x^2)$ → second order in mesh spacing

Thus the difference operator is second-order accurate

→ If f is single-valued, finite and cont. The taylor series for f is convergent
  ↳ This means that as $\Delta x \to \infty$, TE for higher order approx. will be smaller than that of a lower-order TE

→ Similarly, from the Taylor expansion we get:

"central difference" $\frac{f(x+\Delta x, y) - 2f(x,y) + f(x-\Delta x, y)}{\Delta x^2} \approx \frac{\partial^2 f}{\partial x^2}$; $TE = \mathcal{O}(\Delta x^2) = \frac{2\Delta x^2}{4!}\frac{\partial^4 f}{\partial x^4} + \frac{2\Delta x^4}{6!}$

Explicit Method: finding value for n+1, where solution @ n is assumed to be known ← This is the given initial cond. $T(x, t=0) = T_0$

↓
This is just because an algebraic expression

i.e. FDES which use no information at the new time level give uncoupled ODE's

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha\left(\frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2}\right)$$

Implicit Method → Coupled eq. where spatial information about the new time level is needed (but we only know solution at time level n)
↓
Therefore all eq. need to solved simultaneously → $A\vec{x} = \vec{b}$

We could also used a generalized approach of both explicit & implicit → This gives second-order accurate in both space & time

How based on the PDE → There are general requirements for BC's in order to get a solution

↳ hyperbolic, 1st order → need BC only at points where characteristic base curves enter the domain

↳ parabolic / elliptical → BC at every point on the boundary of spatial domain

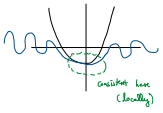
Consistency, Stability & convergence

- Discretization error → replacing PDES w/ FDES
- Stability error → round off errors (finite digits used in computations)
- Truncation error → $\mathcal{O}(\Delta t, \Delta x^2)$ error due to approximation

1) Consistency

$$\lim_{\substack{(\Delta x \to 0) \\ \Delta t \to 0}} (FDE - PDE)_i^n = \lim_{\Delta x \to 0} (TE)_i^n = 0$$

How well does the FDE locally approx the PDE



How to check?

$$\frac{\partial T}{\partial t} = \alpha\left(\frac{\partial^2 T}{\partial x^2}\right)$$

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha\left(\frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2}\right)$$

Expand all terms via Taylor series & then substitute:

$$T_{i+1}^n = T_i^n + \left(\frac{\partial T}{\partial x}\right)_n^i \Delta x + \cdots$$
$$T_{i-1}^n = T_i^n - \left(\frac{\partial T}{\partial x}\right)_n^i \Delta x + \cdots$$
$$T_i^{n+1} = T_i^n + \left(\frac{\partial T}{\partial t}\right)_i^n \Delta t + \cdots$$

↓
Let $\Delta t \to 0$, $\Delta x \to 0$ → if same as PDE, then consistent

↳ Unconditional vs Conditional consistency
↓ no dependence on rate of Δx & Δt ↳ depends on relative rates of Δx & $\Delta t \to 0$

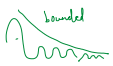
↳ Since Δx & Δt are finite computationally → consistency alone does not guarantee accuracy
there a consistent solve still will have drift

no dependence on ratio of Δx & Δt

depends on relative ratio of Δx & $\Delta t \to 0$

↳ Since Δx & Δt are finite computationally → consistency alone does not guarantee accuracy.
Thus a consistent soln. still behaves diff. than PDE

2) (Numerical) stability :
+ when time domain is involved (i.e. accumulation or growth of errors) (parabolic & hyperbolic PDE's)
+ Depends solely on FDE ; can be bounded (i.e. don't grow)
+ Computational domain must include all of the analytical domain



$\phi = \frac{\alpha \Delta t}{\Delta x^2} \le \frac{1}{2}$ → conditionally stable
↑ dimensionless parameter

Any chosen Δt, Δx → unconditionally stable (most implicit methods)

How to check ?
- Discrete perturbation → perturbation ε @ i & n
- Fourier method → disturbance ε at every grid point at an arbitrary time n
- Matrix method
- Modified equation method

$\left|\frac{\Delta T}{\varepsilon}\right| \le 1$ → perturbations cannot grow
& $\frac{\Delta T}{\varepsilon} > 0$ decay must be monotonic
} Hence need to do for only one time step

- Obtain an evaluation eq: i.e. w/ ε as $f(\Delta x, \Delta t)$
- Use fourier series expansion : $\varepsilon_i^n = \sum_{-\infty}^{\infty} A_j^n e^{Ijhx}$ (fourier component (Amplitude A_j^n, wave. no. = h_j); amplification factor)
- Analyze error in the components $A_j^{n+1} = G_j A_j^n$
- Accounts well for periodic BC (but other BC's may be an issue)

Assume distribution of errors along the x-axis is given by a fourier series in x & time-wise variation is exponential in t
↓
$\varepsilon(x,t) = e^{\alpha t} \sum_m e^{Ikx}$

$G = \left|\frac{\varepsilon_i^{n+1}}{\varepsilon_i^n}\right| \le 1$ [Errors should not be growing]

↳ For $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$
↓
$\frac{\Delta t}{(\Delta x)^2} \le \frac{1}{2}$ [stability requirement]

$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$
$C = c\frac{\Delta t}{\Delta x} \le 1$
↓ courant number
(from von-Neumann stability analysis) [called CFL condition]

FDE's at every interior & boundary grid point are written in matrix form
$T^{n+1} = AT^n + b$
- introduce disturbance to get $\varepsilon^{n+1} = A^{(n+1)}\varepsilon^0$

If A is convergent matrix, e^{n+1} will approach zero in the limit as $n \to \infty$
↓
$\|A\| \le 1$; $\rho(A) \le 1$
↑ smallest norm = spectral radius

3) Convergence : If total error (discretization + stability) → 0 as $\Delta x \to 0$, $\Delta t \to 0$
+ If FDE is convergent, then numerical soln of any accuracy can be obtained by changing the grid-spacing & time-step.
+ FDE must give same solution as PDE in some limiting cases

- Lax-Wendroff method to derive an FDE
  ↳ express time derivative in terms of spatial derivatives
  ↳ Use Taylor series for each point
- Courant no. [$C_o = \frac{c\Delta t}{\Delta x}$]

If $C_o = 1$, then truncation error is zero

For example :

$u_t = K u_{xx}$; $K > 0$; $u(t,x)$

$\frac{(u_i^{n+1} - u_i^n)}{\Delta t}$ ↓ can also be written as

$= \frac{u_i^n(t+\Delta t, x) - u_i^n(t,x)}{\Delta t}$

$= \frac{\left[u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2\right] - u_i^n}{\Delta t}$

Taylor series:
$f(x)\Big|_{x=a} = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 \cdots$
$= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$

$\therefore \quad u_t = \left(\frac{\partial u}{\partial t}\right)_i^n + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \frac{\Delta t}{2} + O(\Delta t^2)$

Similarly $K u_{xx} = \frac{1}{\Delta x^2}\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right)$
($u(t, x+\Delta x)$, $u(t, x-\Delta x)$)

$= \frac{1}{\Delta x^2}\left[\left(u_i^n + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2} + \cdots\right) - 2u_i^n + \left(u_i^n - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2}\right)\right]$

$= \frac{1}{\Delta x^2}\left[\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 + \left(\frac{\partial^4 u}{\partial x^4}\right)_i^n \frac{\Delta x^4}{4!} \cdot 2 + \cdots\right]$

$= \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n + \left(\frac{\partial^4 u}{\partial x^4}\right)_i^n \frac{\Delta x^2}{12} + O(\Delta x^2)$

$\therefore$ Modified equation : $u_t - K u_{xx} = 0$

$\left(\frac{\partial u}{\partial t}\right)_i^n + \frac{\Delta t}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n + O(\Delta t^2) = K\left[u_{xx} + \frac{\Delta x^2}{12} u_{xxxx} + O(\Delta x^4)\right]$

Neglecting higher order terms:

$\frac{\partial u}{\partial t} - K\frac{\partial^2 u}{\partial x^2} = K\frac{\Delta x^2}{12}\frac{\partial^4 u}{\partial x^4} - \underbrace{\frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2}}$

this makes it hyperbolic

tt, t, xx, xxxx